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Lab: Lab 7

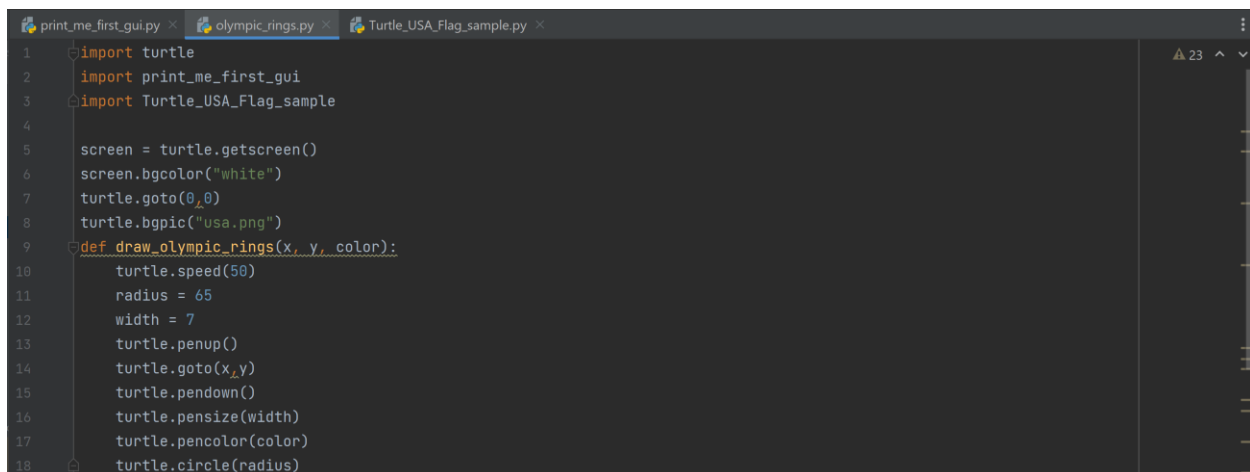
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a. Program Description:

The purpose of program is to use Python's turtle library to create a GUI which shows the map of USA in the background. The program then prints flag of USA, Olympic rings. In addition to that using turtle's write function, the heading for the screen is written as US Olympic team followed by Name, time and program name on bottom right corner of the screen. There are 3 programs used- main program where the Olympic rings are printed, US flag program which is imported in the Olympic Rings program, print me program which is again imported in the Olympic rings program. Source code for all the 3 programs are given below

b. Program Source Code:

Olympic Rings:



```
print_me_first_gui.py x olympic_rings.py x Turtle_USA_Flag_sample.py x
1 import turtle
2 import print_me_first_gui
3 import Turtle_USA_Flag_sample
4
5 screen = turtle.getscreen()
6 screen.bgcolor("white")
7 turtle.goto(0,0)
8 turtle.bgpic("usa.png")
9 def draw_olympic_rings(x, y, color):
10     turtle.speed(50)
11     radius = 65
12     width = 7
13     turtle.penup()
14     turtle.goto(x,y)
15     turtle.pendown()
16     turtle.pensize(width)
17     turtle.pencolor(color)
18     turtle.circle(radius)
```

```
print_me_first_gui.py × olympic_rings.py × Turtle_USA_Flag_sample.py ×
19
20 def write_info():
21     turtle.penup()
22     width = 100
23     turtle.pensize(width)
24     turtle.goto(0, 150)
25     turtle.pendown()
26     turtle.pencolor("blue")
27     style = ('Ariel', 70, 'bold')
28     turtle.write("USA\n", font=style, align='center')
29     turtle.write("OLYMPIC TEAM", font=style, align='center')
30     #turtle.write("OLYMPIC TEAM", font=style, align='center')
31     turtle.hideturtle()
32
33 def write_name():
34     width = 10
35     turtle.penup()
36     turtle.pensize()
37     turtle.goto(300,-300)
38     turtle.pendown()
39     style = ('Ariel', 10, 'normal')
40     infoDict = print_me_first_gui.print_me_time()
41     turtle.write(infoDict[0], font=style)
42     turtle.penup()
43     turtle.goto(300, -315)
44     turtle.pendown()
45     turtle.write(infoDict[1], font=style)
46     turtle.penup()
47     turtle.goto(300, -330)
```

```
print_me_first_gui.py × olympic_rings.py × Turtle_USA_Flag_sample.py ×
33 def write_name():
34     width = 10
35     turtle.penup()
36     turtle.pensize()
37     turtle.goto(300,-300)
38     turtle.pendown()
39     style = ('Ariel', 10, 'normal')
40     infoDict = print_me_first_gui.print_me_time()
41     turtle.write(infoDict[0], font=style)
42     turtle.penup()
43     turtle.goto(300, -315)
44     turtle.pendown()
45     turtle.write(infoDict[1], font=style)
46     turtle.penup()
47     turtle.goto(300, -330)
48     turtle.pendown()
49     turtle.write(infoDict[2], font=style)
50     turtle.hideturtle()
51
52 Turtle_USA_Flag_sample.draw_US_flag()
53 draw_olympic_rings(-160,-300, "blue")
54 draw_olympic_rings(-80,-360, "orange")
55 draw_olympic_rings(0,-300, "black")
56 draw_olympic_rings(80,-360, "green")
57 draw_olympic_rings(160,-300, "red")
58 write_info()
59 print_me_first_gui.print_me_time()
60 screen.mainloop()
61
```

Print Name(print_me_first_gui.py):

```
print_me_first_gui.py x olympic_rings.py x Turtle_USA_Flag_sample.py x
1 import os.path
2 import sys
3 from datetime import datetime
4 import turtle
5
6 def print_me_time():
7     name = "Shweta Chauhan"
8     program = os.path.basename((sys.argv[0]))
9     currentTime = datetime.now()
10    timestamp_str = currentTime.strftime("%Y-%m-%d %H:%M:%S")
11    turtle.pensize()
12    position(300, -300)
13    write("Name")
14    position(380, -300)
15    write(": " + name)
16    position(300, -320)
17    write("Program")
18    position(380, -320)
19    write(": " + program)
20    position(300, -340)
21    write("Time")
22    position(380, -340)
23    write(": " + timestamp_str)
24
25 def position(x,y):
26     turtle.penup()
27     turtle.goto(x,y)
28     turtle.pendown()
29
30 def write(info):
31     style = ('Ariel', 10, 'normal')
32     turtle.write(info, font=_style)
33
34
35
36
37
38
```

c. Program Output

